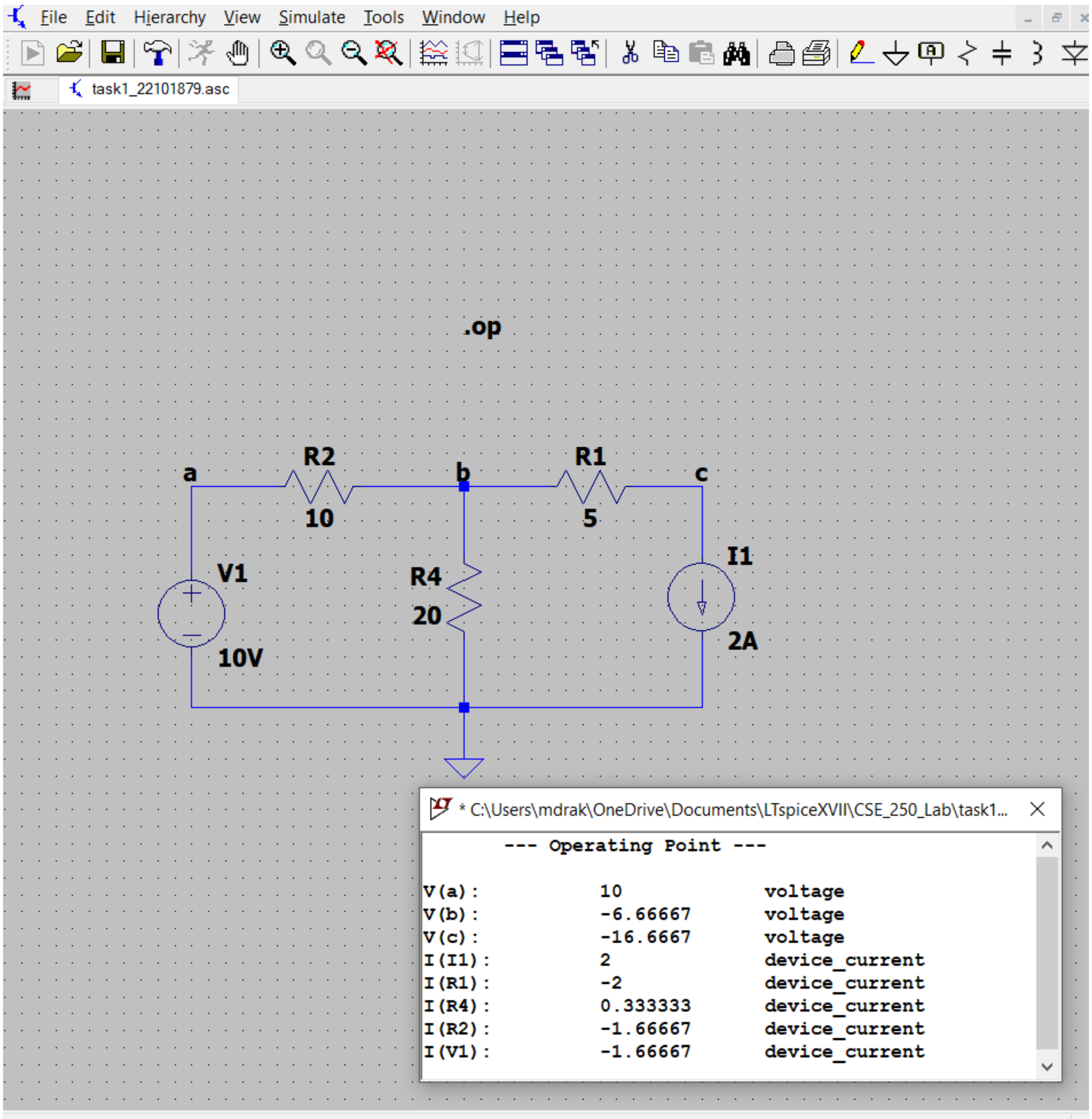
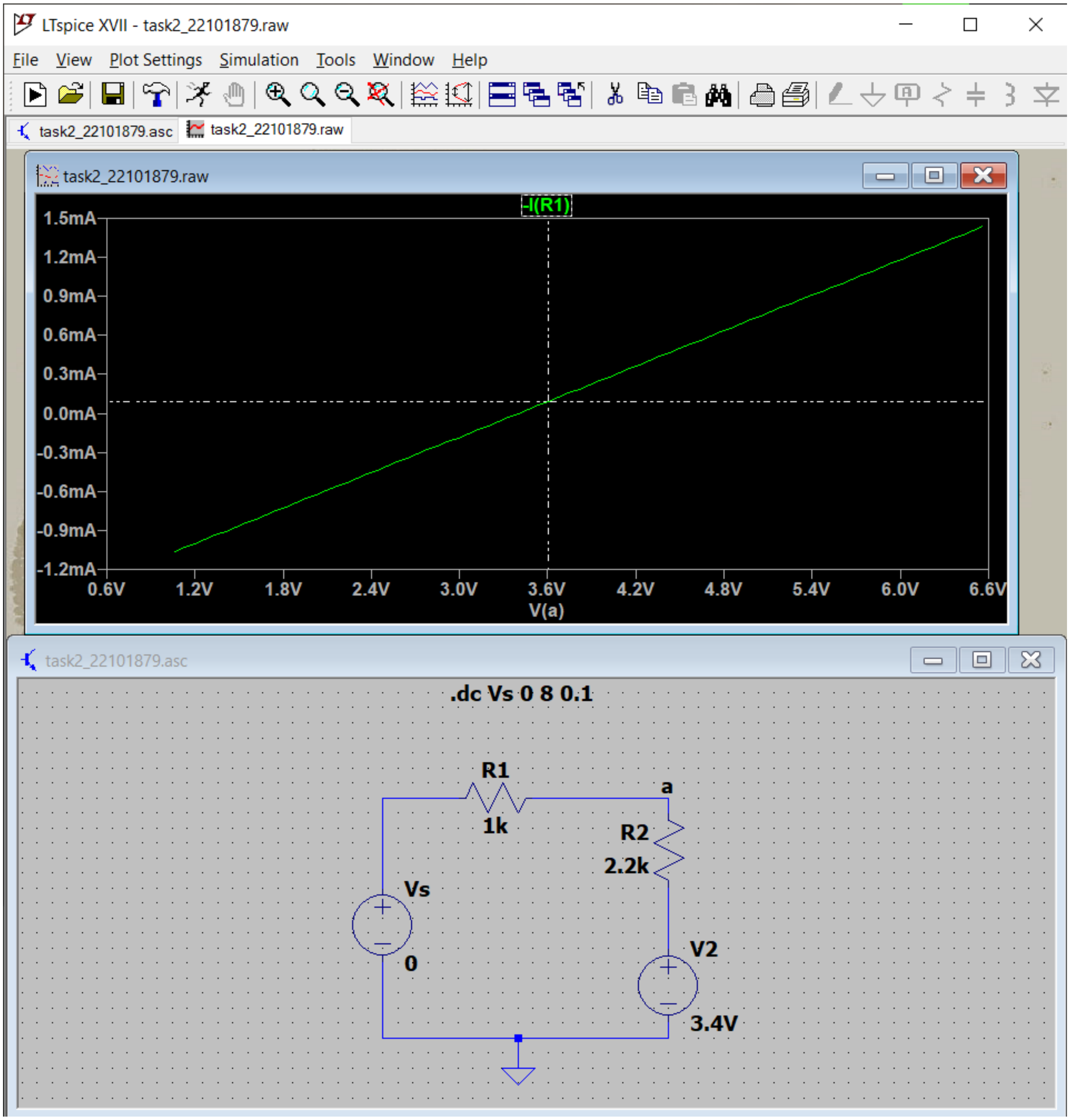
Name: MD Rakib Hossain Ontu
Id:22101879
Sec: 16
Group: 04

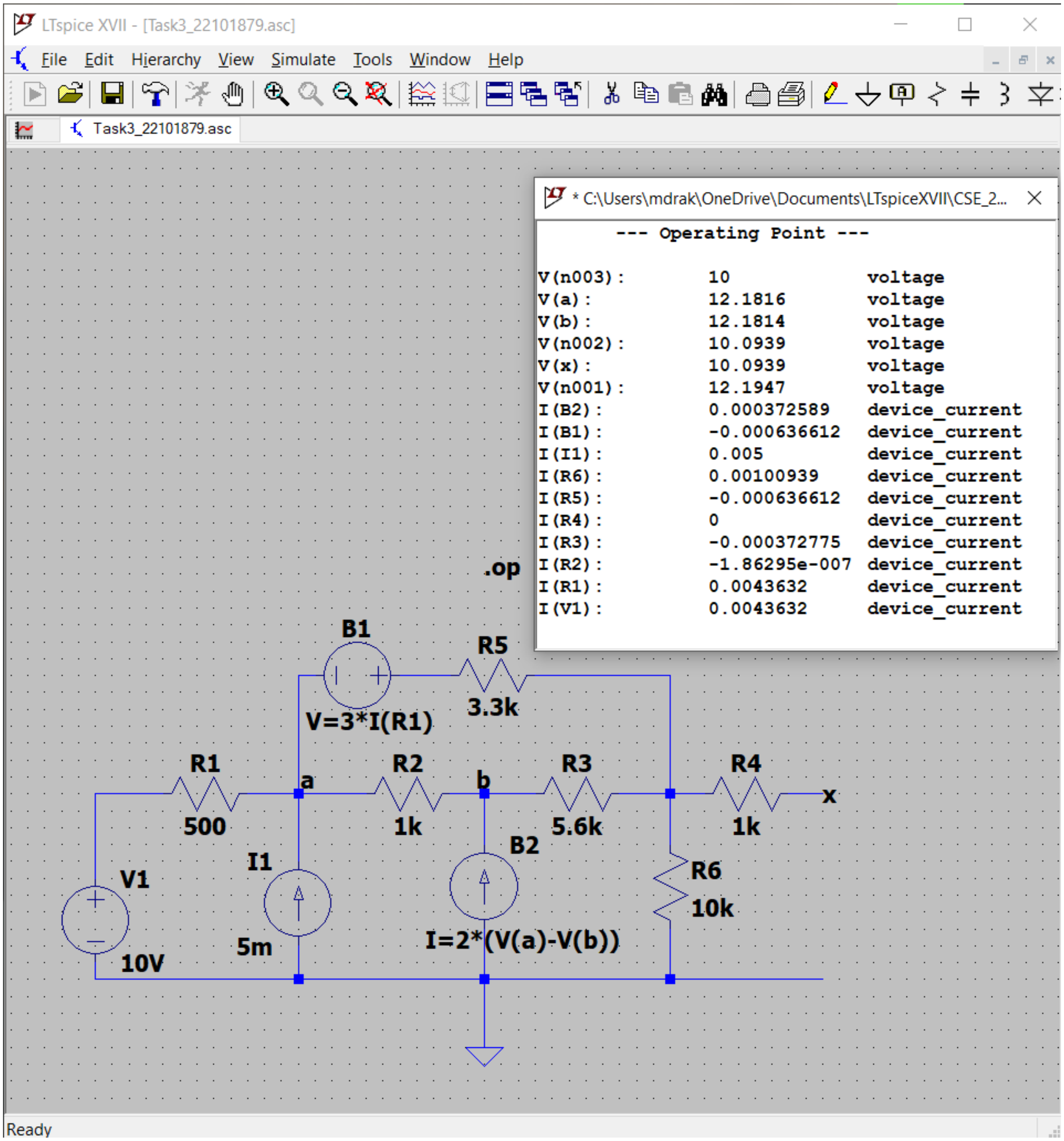
task1



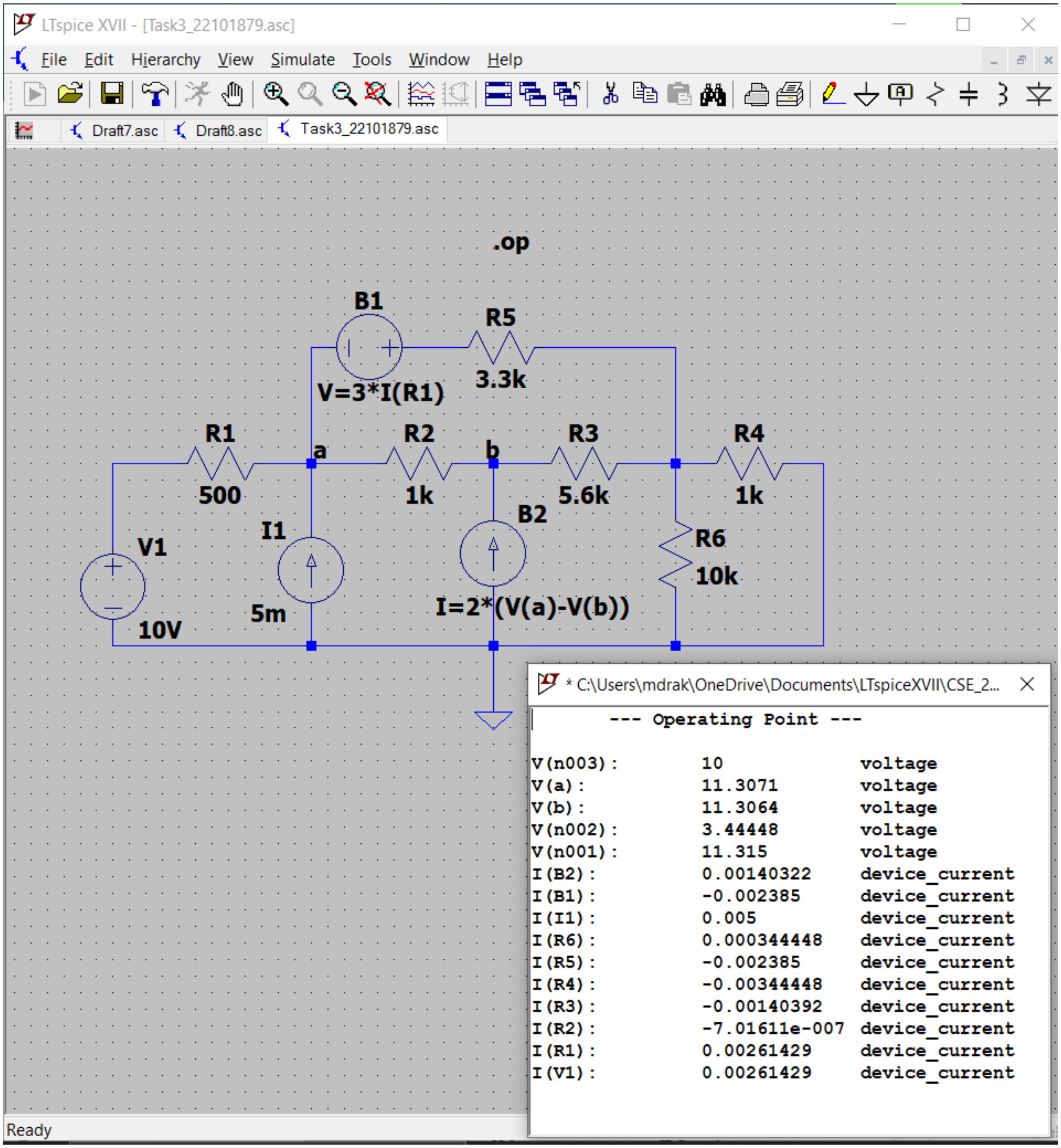
Task2


LTspice XVII - task2_22101879.raw
File View Plot Settings Simulation Tools Window Help
task2_22101879.asc
task2_22101879.raw
task2_22101879.raw
-I(R1)
1.5mA
1.2mA
0.9mA
0.6mA
0.3mA
0.0mA
-0.3mA
-0.6mA
-0.9mA
-1.2mA
0.6V
1.2V
1.8V
2.4V
3.0V
3.6V
4.2V
4.8V
5.4V
6.0V
6.6V
V(a)
task2_22101879.asc
.dc Vs 0 8 0.1
R1
1k
a
R2
2.2k
Vs
0
V2
3.4V

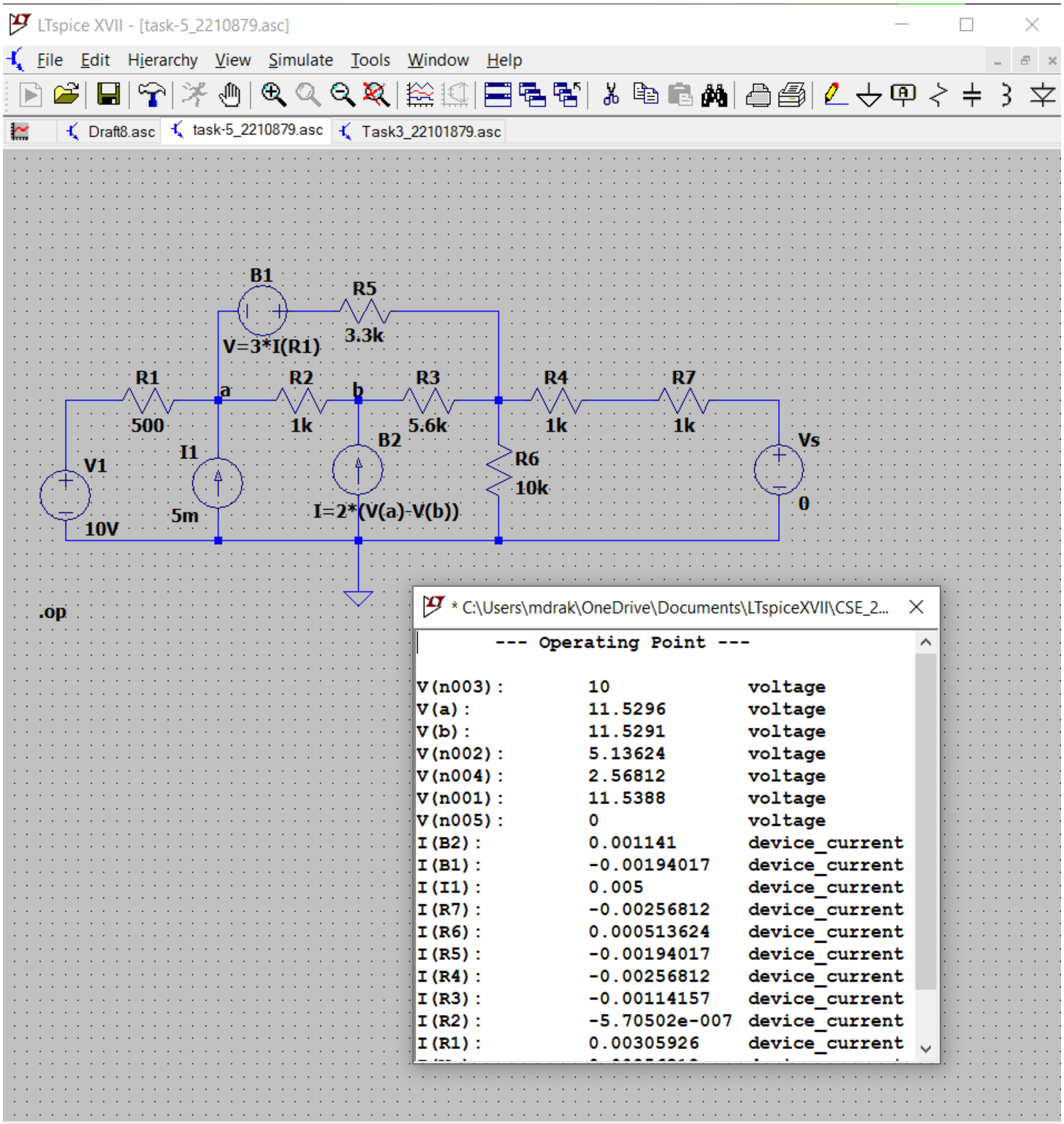
Task3.1


LTspice XVII - [Task3_22101879.asc]
File Edit Hierarchy View Simulate Tools Window Help
Task3_22101879.asc
* C:\Users\mdrak\OneDrive\Documents\LTspiceXVII\CSE_2...
--- Operating Point ---
V(n003): 10 voltage
V(a): 12.1816 voltage
V(b): 12.1814 voltage
V(n002): 10.0939 voltage
V(x): 10.0939 voltage
V(n001): 12.1947 voltage
I(B2): 0.000372589 device_current
I(B1): -0.000636612 device_current
I(I1): 0.005 device_current
I(R6): 0.00100939 device_current
I(R5): -0.000636612 device_current
I(R4): 0 device_current
I(R3): -0.000372775 device_current
I(R2): -1.86295e-007 device_current
I(R1): 0.0043632 device_current
I(V1): 0.0043632 device_current
.op
B1
V=3*I(R1)
R5
3.3k
R1
500
a
R2
1k
b
R3
5.6k
R4
1k
x
B2
R6
10k
I1
5m
V1
10V
I=2*(V(a)-V(b))
Ready

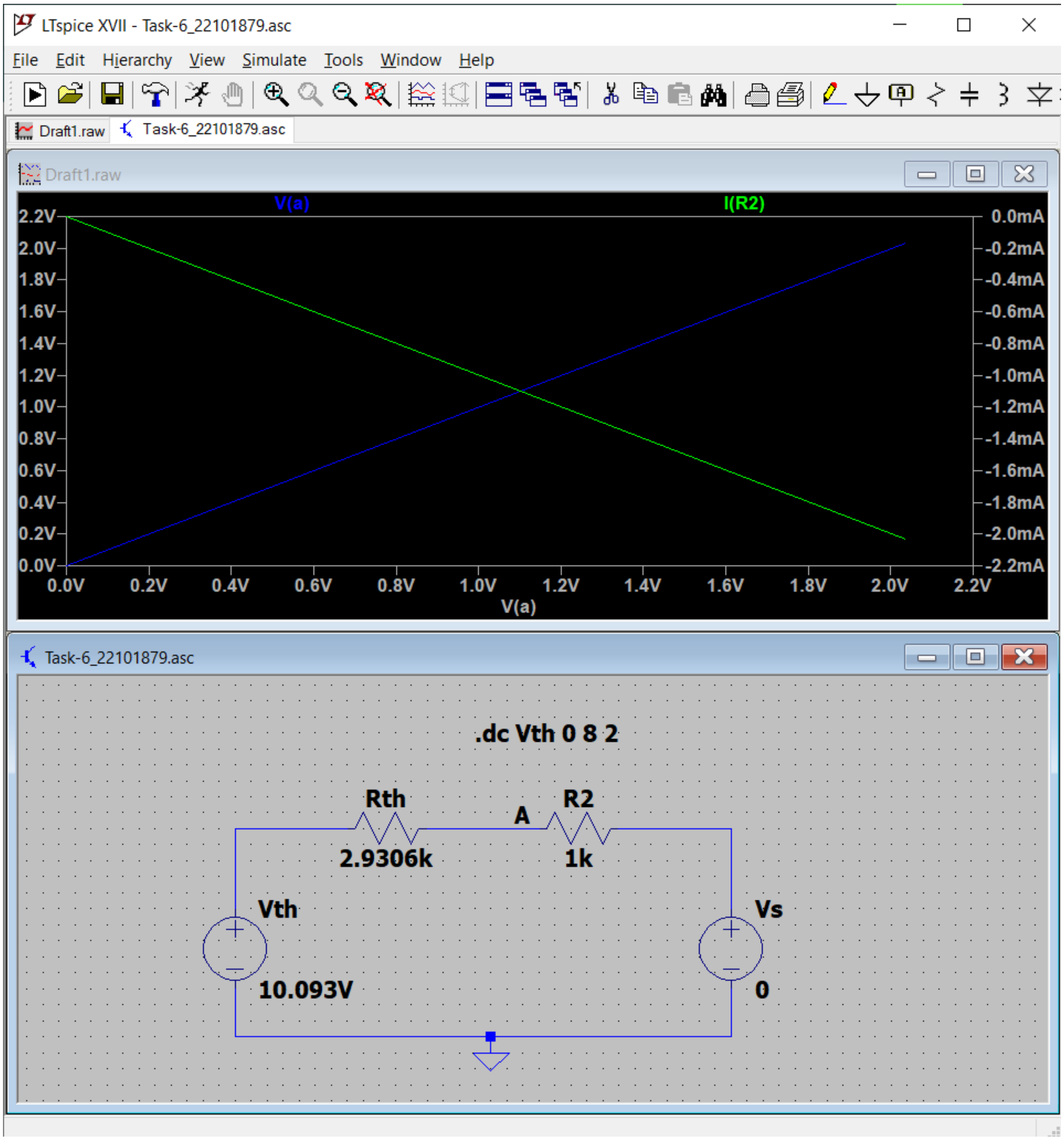
Task-3.2


LTspice XVII - [Task3_22101879.asc]
File Edit Hierarchy View Simulate Tools Window Help
Draft7.asc
Draft8.asc
Task3_22101879.asc
.op
B1
V=3*I(R1)
R5
3.3k
R1
500
R2
1k
R3
5.6k
R4
1k
a
b
V1
10V
I1
5m
B2
I=2*(V(a)-V(b))
R6
10k
Ready
* C:\Users\mdrak\OneDrive\Documents\LTspiceXVII\CSE_2...
--- Operating Point ---
V(n003): 10 voltage
V(a): 11.3071 voltage
V(b): 11.3064 voltage
V(n002): 3.44448 voltage
V(n001): 11.315 voltage
I(B2): 0.00140322 device_current
I(B1): -0.002385 device_current
I(I1): 0.005 device_current
I(R6): 0.000344448 device_current
I(R5): -0.002385 device_current
I(R4): -0.00344448 device_current
I(R3): -0.00140392 device_current
I(R2): -7.01611e-007 device_current
I(R1): 0.00261429 device_current
I(V1): 0.00261429 device_current

Task-4

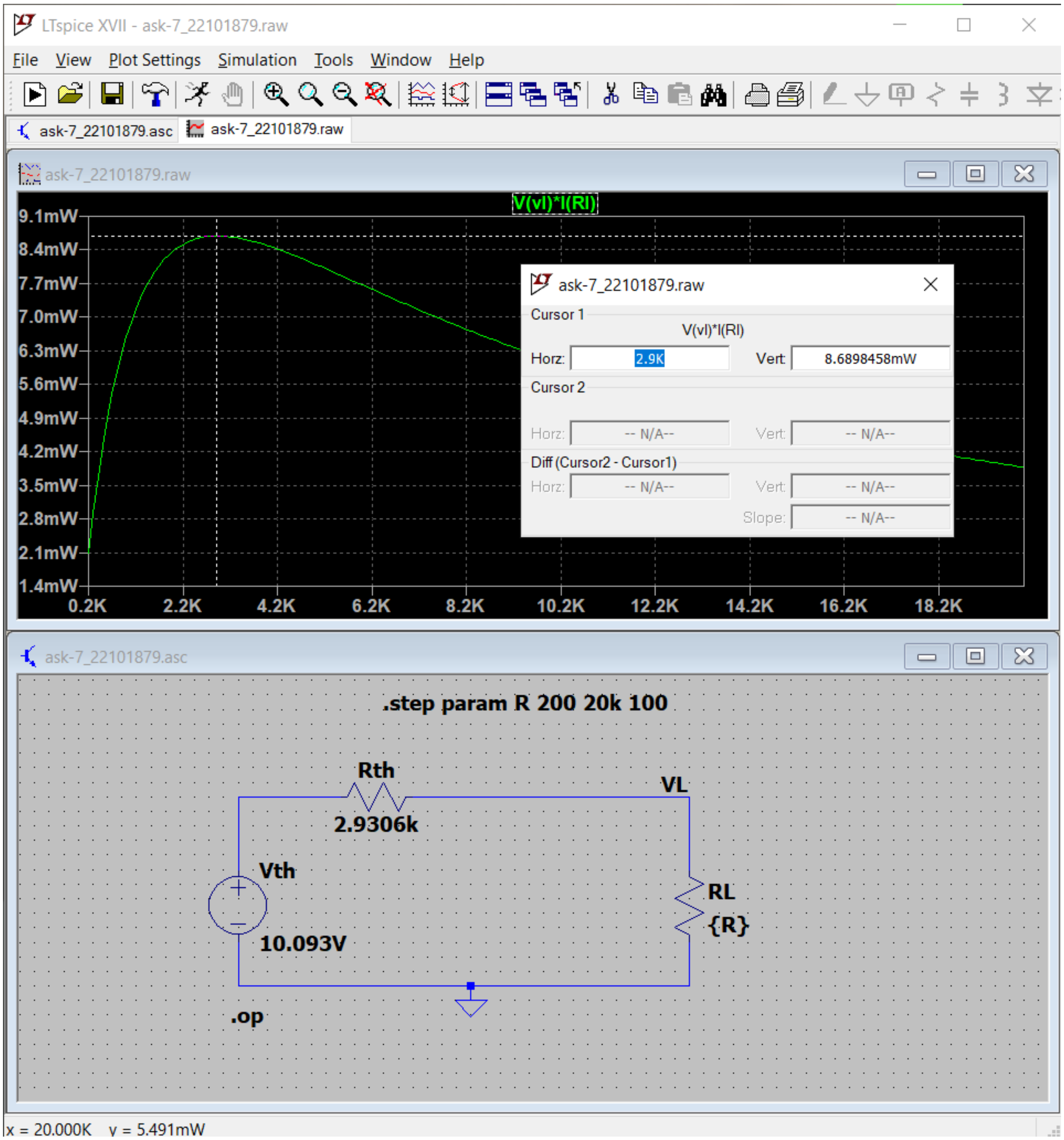

LTspice XVII - [task-5_2210879.asc]
File Edit Hierarchy View Simulate Tools Window Help
Draft8.asc
task-5_2210879.asc
Task3_22101879.asc
B1
V=3*I(R1)
R5
3.3k
R1
500
a
R2
1k
b
R3
5.6k
R4
1k
R7
1k
V1
10V
I1
5m
B2
I=2*(V(a)-V(b))
R6
10k
Vs
0
.op
* C:\Users\mdrak\OneDrive\Documents\LTspiceXVII\CSE_2...
--- Operating Point ---
V(n003): 10 voltage
V(a): 11.5296 voltage
V(b): 11.5291 voltage
V(n002): 5.13624 voltage
V(n004): 2.56812 voltage
V(n001): 11.5388 voltage
V(n005): 0 voltage
I(B2): 0.001141 device_current
I(B1): -0.00194017 device_current
I(I1): 0.005 device_current
I(R7): -0.00256812 device_current
I(R6): 0.000513624 device_current
I(R5): -0.00194017 device_current
I(R4): -0.00256812 device_current
I(R3): -0.00114157 device_current
I(R2): -5.70502e-007 device_current
I(R1): 0.00305926 device_current

Task-6:



Answer: The given circuit does follow the Thevenin's equivalent circuit, as we know in the Thevenin's theorem the source of voltage Rth is the equivalent resistance of the original circuit as viewed from the output terminals when all sources are off, and Vth is the open-circuit voltage at the original circuit's output terminals.Since the given circuit does the same, so we can say it follows the theorem.

Task:7



Answer:
From the above plot we can see that the maximum power of Rl was 8.898458mW at 2.906. So maximum power had been achieved at Rth register. Hence the circuit follows Thevenin's theorem.